Brno University of Technology
Faculty of Information Technology

IVS Project – (INTERCALCULATOR)

VERSION 1.0.1

# Programmer's Manual

**Author:**
Ha Pham (xphamha00)

Current as of April 28, 2026

# Contents

**1 Current Appearance of the Calculator** **2**

**2 Introduction** **3**

2.0.1 Getting Started . . . . . . . . . . . . . . . . . . . . . . . . . . . . 3

**3 Prerequisites** **3**

**4 Architecture** **3**

**5 Testing** **4**

**6 Makefile and Automation** **4**

# 1 Current Appearance of the Calculator

# 2 Introduction

This document serves as a supplementary manual for developers working on the `INTERCALCULATOR` project. Reading the entire manual is strongly recommended. This document is not intended for end users of the application.

### 2.0.1 Getting Started

1) It is recommended to clone the project into a chosen target directory:

```
$ git clone https://github.com/honzakralem/ivs-project-2
```

2) Create a virtual environment using the make command:

```
$ make
```

3) Activate the virtual environment. This will differ depending on the operating system. For Linux/macOS:

```
$ source env/bin/activate
```

For Windows:

```
$ env\Scripts\activate
```

4) It is strongly recommended to configure your git name and email settings (although it is not mandatory):

```
$ git config --global user.name "name"
$ git config --global user.email "email"
```

# 3 Prerequisites

For the current phase of development, the following tools/programs must be installed:

- `PYTHON3:` required for running the code itself
- `GNU Make:` required for running automation scripts
- `GIT:` required for version control and collaboration with other developers via the remote repository
- `DOXYGEN:` used for generating documentation
- `GITHUB CLI:` used for downloading the latest installers from Releases for make pack

# 4 Architecture

This section describes the logic and structure of the project, including files and directories along with their functions.

- `src` - source code
  - `mathlib.py` - implementation of mathematical functions

  - `mathlib_test.py` - unit tests that verify the correct functionality of mathlib.py
  - `requirements.txt` - list of external libraries required by the project to run or be tested
  - `infixtopost.py` - converts a string from infix notation (e.g. $2 + 2$) to postfix notation (e.g. $2\ 2\ +$)
  - `infixtopost_test.py` - unit tests for the functionality of infixtopost.py functions
  - `gui.py` - main graphical user interface
  - `setup.py` - script used to install dependencies
  - `Doxyfile` - configuration file for *Doxygen* documentation
  - The source directory also contains the main `makefile` and `.gitignore`
- `mockup` - directory containing the future user interface design of the calculator
- `plan` - directory containing the project plan and deadlines
- `profiling` - directory containing programs for measuring calculator performance
- The main project directory will contain `LICENSE`, `README.MD`, and `.gitignore`

# 5 Testing

This section covers testing the functionality of the modules. To run the tests, simply execute the following command from the `src` folder:

```
$ make test
```

This will run `infixtopost_test.py` and `mathlib_test.py`. The results will be reported with a status message.

# 6 Makefile and Automation

This section primarily covers the `makefile` located in the `src` directory. It contains commands for automating development tasks. All available commands can be listed (assuming the current working directory is `src`) by running:

```
$ make help
```